```
import numpy as np
import pandas as pd
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
from sklearn.linear_model import Lasso
from sklearn import metrics
from sklearn.preprocessing import StandardScaler
import datetime
import matplotlib.pyplot as plt
%matplotlib inline
import warnings
warnings.filterwarnings('ignore')
```

```
car_dataset = pd.read_csv('/content/car details v4.csv')
```

```
car_dataset.head()
```

	Make	Model	Price	Year	Kilometer	Fuel_Type	Transmission	Location	Color	Owner	Seller_Type	Engine	Max_Power
0	Honda	Amaze 1.2 VX i-VTEC	505000	2017	87150	Petrol	Manual	Pune	Grey	First	Corporate	1198 cc	87 bhp @ 6000 rpm
1	Maruti Suzuki	Swift DZire VDI	450000	2014	75000	Diesel	Manual	Ludhiana	White	Second	Individual	1248 cc	74 bhp @ 4000 rpm
2	Hyundai	i10 Magna 1.2 Kappa2	220000	2011	67000	Petrol	Manual	Lucknow	Maroon	First	Individual	1197 cc	79 bhp @ 6000 rpm
3	Toyota	Glanza G	799000	2019	37500	Petrol	Manual	Mangalore	Red	First	Individual	1197 cc	82 bhp @ 6000 rpm
4	Toyota	Innova 2.4 VX 7 STR [2016-2020]	1950000	2018	69000	Diesel	Manual	Mumbai	Grey	First	Individual	2393 cc	148 bhp @ 3400 rpm

Next steps: Generate code with car_dataset New interactive sheet

```
car_dataset.isnull().sum()
```

	0
Make	0
Model	0
Price	0
Year	0
Kilometer	0
Fuel_Type	0
Transmission	0
Location	0
Color	0
Owner	0
Seller_Type	0
Engine	80
Max_Power	80
Max_Torque	80
Drivetrain	136
Length	64
Width	64
Height	64
Seating_Capacity	64
Fuel_Tank_Capacity	113

dtype: int64

```
car_dataset.shape
```

```
(2059, 20)
```

```
car_dataset['Engine'] = car_dataset['Engine'].astype(str).str.extract(r'(\d+\.?\d*)')[0]
car_dataset['Max_Power'] = car_dataset['Max_Power'].astype(str).str.extract(r'(\d+\.?\d*)')[0]
car_dataset['Max_Torque'] = car_dataset['Max_Torque'].astype(str).str.extract(r'(\d+\.?\d*)')[0]

car_dataset['Engine'] = pd.to_numeric(car_dataset['Engine'], errors='coerce')
car_dataset['Max_Power'] = pd.to_numeric(car_dataset['Max_Power'], errors='coerce')
car_dataset['Max_Torque'] = pd.to_numeric(car_dataset['Max_Torque'], errors='coerce')
```

```
num_cols = [
    'Engine', 'Max_Power', 'Max_Torque',
    'Length', 'Width', 'Height',
    'Seating_Capacity', 'Fuel_Tank_Capacity'
]

for col in num_cols:
    car_dataset[col] = car_dataset[col].fillna(car_dataset[col].median())

car_dataset['Drivetrain'] = car_dataset['Drivetrain'].fillna(car_dataset['Drivetrain'].mode()[0])

car_dataset.isnull().sum()
```

	0
Make	0
Model	0
Price	0
Year	0
Kilometer	0
Fuel_Type	0
Transmission	0
Location	0
Color	0
Owner	0
Seller_Type	0
Engine	0
Max_Power	0
Max_Torque	0
Drivetrain	0
Length	0
Width	0
Height	0
Seating_Capacity	0
Fuel_Tank_Capacity	0

dtype: int64

```
current_year = datetime.datetime.now().year
car_dataset['Car_Age'] = current_year - car_dataset['Year']
```

```
car_dataset.head()
```

	Make	Model	Price	Year	Kilometer	Fuel_Type	Transmission	Location	Color	Owner	Seller_Type	Engine	Max_Power
0	Honda	Amaze 1.2 VX i-VTEC	505000	2017	87150	Petrol	Manual	Pune	Grey	First	Corporate	1198.0	87.0
1	Maruti Suzuki	Swift DZire VDI	450000	2014	75000	Diesel	Manual	Ludhiana	White	Second	Individual	1248.0	74.0
2	Hyundai	i10 Magna 1.2 Kappa2	220000	2011	67000	Petrol	Manual	Lucknow	Maroon	First	Individual	1197.0	79.0
3	Toyota	Glanza G	799000	2019	37500	Petrol	Manual	Mangalore	Red	First	Individual	1197.0	82.0
4	Toyota	Innova 2.4 VX 7 STR [2016-2020]	1950000	2018	69000	Diesel	Manual	Mumbai	Grey	First	Individual	2393.0	148.0

```
car_dataset['Fuel_Type'].value_counts()
```

Fuel_Type	count
Diesel	1049
Petrol	942
CNG	50
Electric	7
LPG	5
Hybrid	3
CNG + CNG	1
Petrol + CNG	1
Petrol + LPG	1

dtype: int64

```
car_dataset['Owner'].value_counts()
```

Owner	count
First	1619
Second	373
Third	42
UnRegistered Car	21
Fourth	3
4 or More	1

dtype: int64

```
car_dataset['Fuel_Type'] = car_dataset['Fuel_Type'].replace({
    'CNG + CNG': 'CNG',
    'Petrol + CNG': 'Other',
    'Petrol + LPG': 'Other',
    'LPG': 'Other',
    'Hybrid': 'Other'
})

car_dataset['Owner'] = car_dataset['Owner'].replace({
    '4 or More': 'Third',
    'Fourth': 'Third'
})
```

```
car_dataset['Location'].value_counts()
```

Location	count
Mumbai	342
Delhi	307
Pune	144
Bangalore	132
Hyderabad	116
...	...
Valsad	1
Rohtak	1
Faizabad	1
Pimpri-Chinchwad	1
Siliguri	1

77 rows × 1 columns

dtype: int64

```
top_locations = car_dataset['Location'].value_counts().nlargest(10).index
car_dataset['Location'] = car_dataset['Location'].apply(
    lambda x: x if x in top_locations else 'Other'
)
```

```
car_dataset['Location'].value_counts()
```

Location	count
Other	695
Mumbai	342
Delhi	307
Pune	144
Bangalore	132
Hyderabad	116
Lucknow	78
Ahmedabad	70
Chennai	63
Kolkata	60
Kanpur	52

dtype: int64

```
car_dataset = car_dataset.drop(columns=['Color', 'Year'])
```

```
car_dataset.head()
```

	Make	Model	Price	Kilometer	Fuel_Type	Transmission	Location	Owner	Seller_Type	Engine	Max_Power	Max_Torque	Dr
0	Honda	Amaze 1.2 VX i-VTEC	505000	87150	Petrol	Manual	Pune	First	Corporate	1198.0	87.0	109.0000	
1	Maruti Suzuki	Swift DZire VDI	450000	75000	Diesel	Manual	Other	Second	Individual	1248.0	74.0	190.0000	
2	Hyundai	i10 Magna 1.2 Kappa2	220000	67000	Petrol	Manual	Lucknow	First	Individual	1197.0	79.0	112.7619	
3	Toyota	Glanza G	799000	37500	Petrol	Manual	Other	First	Individual	1197.0	82.0	113.0000	
4	Toyota	Innova 2.4 VX 7 STR [2016-2020]	1950000	69000	Diesel	Manual	Mumbai	First	Individual	2393.0	148.0	343.0000	

Next steps: Generate code with car_dataset New interactive sheet

```
X = car_dataset.drop(columns=['Price', 'Make', 'Model'])
Y = np.log(car_dataset['Price'])
```

```
X = pd.get_dummies(X, drop_first=True)
X.head()
```

	Kilometer	Engine	Max_Power	Max_Torque	Length	Width	Height	Seating_Capacity	Fuel_Tank_Capacity	Car_Age	Fuel_Type_Die
0	87150	1198.0	87.0	109.0000	3990.0	1680.0	1505.0	5.0	35.0	9	Fa
1	75000	1248.0	74.0	190.0000	3995.0	1695.0	1555.0	5.0	42.0	12	T
2	67000	1197.0	79.0	112.7619	3585.0	1595.0	1550.0	5.0	35.0	15	Fa
3	37500	1197.0	82.0	113.0000	3995.0	1745.0	1510.0	5.0	37.0	7	Fa
4	69000	2393.0	148.0	343.0000	4735.0	1830.0	1795.0	7.0	55.0	8	T

```
X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
linr_reg_model = LinearRegression()
linr_reg_model.fit(X_train, Y_train)

training_reg_prediction = linr_reg_model.predict(X_train)
```

```
lr_r2 = metrics.r2_score(Y_train, training_reg_prediction)
lr_mae = mean_absolute_error(Y_train, training_reg_prediction)
lr_mse = mean_squared_error(Y_train, training_reg_prediction)
lr_rmse = np.sqrt(mean_squared_error(Y_train, training_reg_prediction))

print("Linear Regression Results on Training Data")
print("Training R2 Score :", lr_r2)
print("Training MAE      :", lr_mae)
print("Training RMSE     :", lr_rmse)
print("Training MSE       :", lr_mse)
```

```
Linear Regression Results on Training Data
Training R2 Score : 0.9086944808697266
Training MAE      : 0.21021064887310617
```

```
Training RMSE     : 0.2970618248147858
Training MSE       : 0.08824572776229052
```

```
# Visualize the actual prices and predicted prices

plt.figure(figsize=(10, 6))
plt.scatter(Y_train, training_reg_prediction, alpha=0.5)
plt.xlabel('Actual Prices')
plt.ylabel('Predicted Prices')
plt.title('Actual Prices vs. Predicted Prices')
plt.show()
```

```
test_reg_prediction = linr_reg_model.predict(X_test)
```

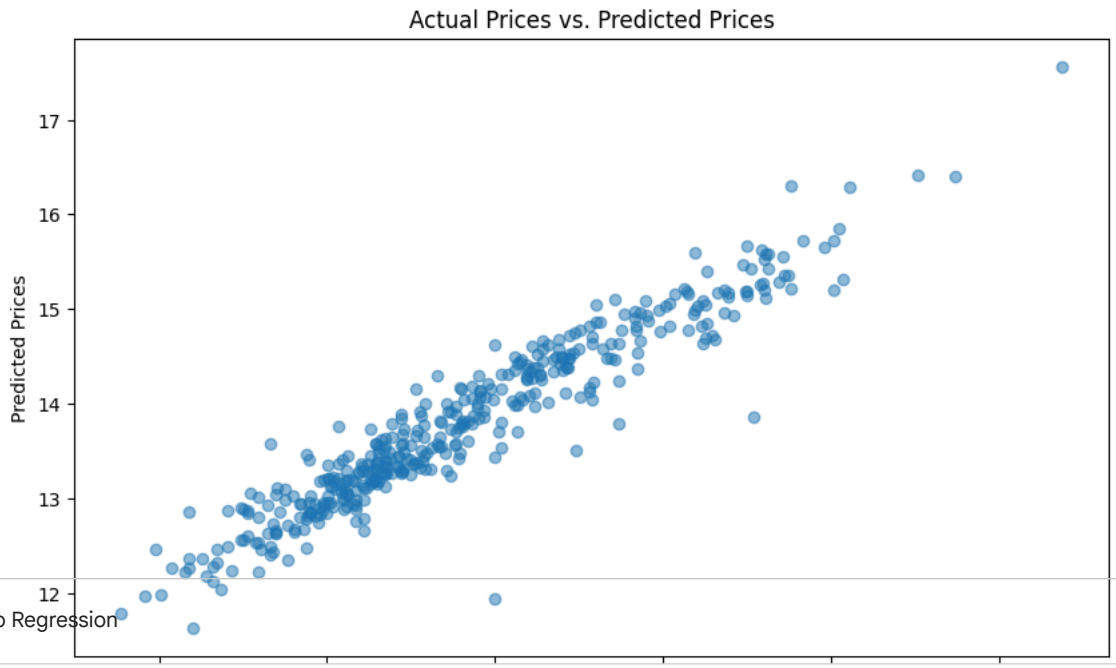
```
lr_r2 = metrics.r2_score(Y_test, test_reg_prediction)
lr_mae = mean_absolute_error(Y_test, test_reg_prediction)
lr_mse = mean_squared_error(Y_test, test_reg_prediction)
lr_rmse = np.sqrt(mean_squared_error(Y_test, test_reg_prediction))

print("Linear Regression Results on Test data")
print("Test R2 Score :", lr_r2)
print("Test MAE      :", lr_mae)
print("Test RMSE     :", lr_rmse)
print("Test MSE       :", lr_mse)
```

```
Linear Regression Results on Test data
Test R2 Score : 0.9099175354942546
Test MAE      : 0.20813431368684798
Test RMSE     : 0.2913772826417529
Test MSE       : 0.08490072083969194
```

```
# Visualize the actual prices and predicted prices

plt.figure(figsize=(10, 6))
plt.scatter(Y_test, test_reg_prediction, alpha=0.5)
plt.xlabel('Actual Prices')
plt.ylabel('Predicted Prices')
plt.title('Actual Prices vs. Predicted Prices')
plt.show()
```



Lasso Regression

```
lasso_reg_model = Lasso(alpha=0.01)
lasso_reg_model.fit(X_train, Y_train)

training_reg_prediction = lasso_reg_model.predict(X_train)
```

```
lr_r2 = metrics.r2_score(Y_train, training_reg_prediction)
lr_mae = mean_absolute_error(Y_train, training_reg_prediction)
lr_mse = mean_squared_error(Y_train, training_reg_prediction)
lr_rmse = np.sqrt(mean_squared_error(Y_train, training_reg_prediction))

print("Lasso Regression Results on Training Data")
print("Training R2 Score :", lr_r2)
print("Training MAE      :", lr_mae)
print("Training RMSE     :", lr_rmse)
print("Training MSE       :", lr_mse)
```

```
Lasso Regression Results on Training Data
Training R2 Score : 0.9062837217081721
```